

Voice Interview Coach

An offline voice interviewer

A mock technical interview you talk to, running entirely on one laptop – speech recognition, a local language model and speech synthesis, with every sentence checked against a contract before it is spoken.

1.25 s

median to first audio

181

backend tests

95.1%

coverage, lines and branches

21

contract rules, checked per sentence

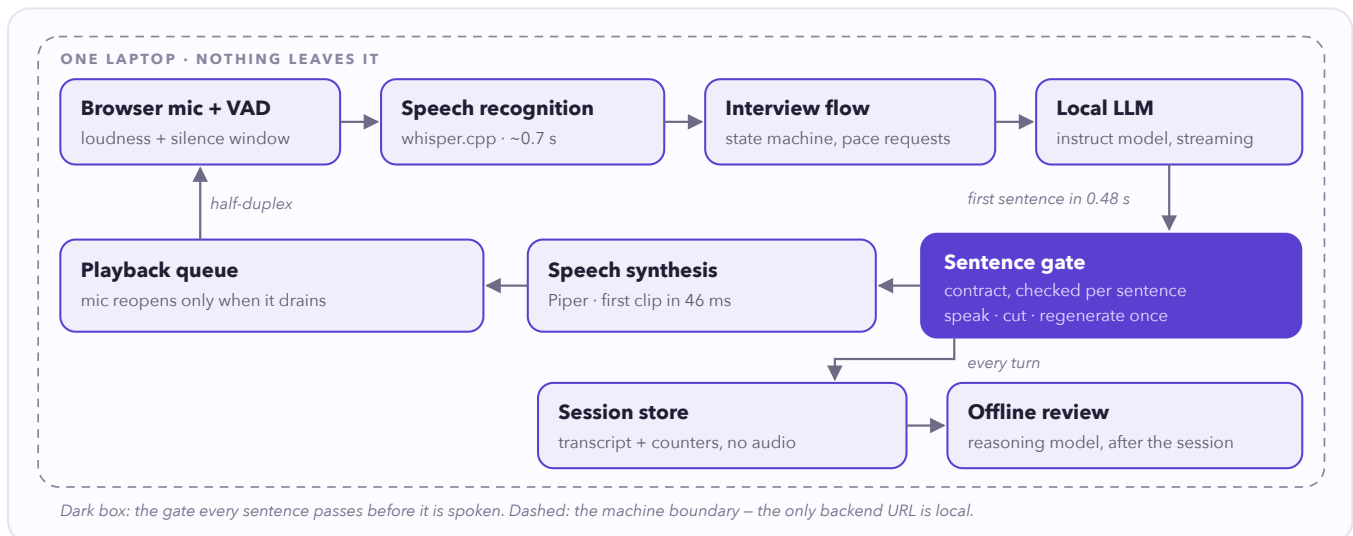
THE PROBLEM

A voice turn feels live only if speech starts within seconds of the user stopping – and that window must hold end-of-turn detection, recognition, a large model's first sentence and synthesis on one 32 GB laptop, with nothing sent to a cloud. The shortcuts fail on the numbers: a buffered reply projects to 4.8 s at p95, and a reasoning model took 31–52 s.

WHAT IT DOES

- Runs a 15-minute English interview: six questions, technical and behavioral
- Transcribes locally and answers aloud, sentence by sentence
- Checks every sentence against a contract before it is voiced
- Writes the review afterwards with a separate reasoning model

ARCHITECTURE



HOW IT FLOWS

- 1 The browser detects the end of your turn; the utterance is transcribed locally.
- 2 The model streams its reply; each sentence is checked before it reaches synthesis.
- 3 Valid sentences play back to back; the mic reopens only when playback drains.
- 4 After the session, a reasoning model writes the review, validated against a schema.

WORTH KNOWING

The model may agree to say it; you never hear it.

A bad first sentence is regenerated before any sound; a bad later sentence is cut before synthesis.

A reasoning model meant 31–52 s of silence.

So it writes only the offline review; the live voice runs on an instruct model at 1.25 s.

“Speak slower” never reaches the LLM.

The coach confirms and repeats its last line at the new pace – without spending a question.

DECISIONS THAT SHAPE IT

Split the models: instruct live, reasoning offline

01

Because the reasoning model took 31-52 s per reply against 0.19 s to the first token for the instruct model. Depth moved to the review, where nobody waits.

Cost: the live dialogue loses the reasoning model's depth.

Stream sentence by sentence, not a buffered reply

02

Because a buffered reply projected to 4.8 s at p95 against 3.0 s streamed; the measured median to first audio is 1.25 s.

Cost: the contract never sees the whole reply, and pauses are written into the audio itself.

Check each sentence before it is spoken

03

Because the checks are monotone, so what has been said is always compliant. Without recovery, compliance decays: 0.95 per turn leaves 0.36 after 20 turns.

Cost: only word-level checks; tone and persona stay in the prompt.

Half-duplex turn-taking

04

Because an open mic during playback made the coach interrupt itself. The browser detects the end of a turn; the mic reopens only when playback drains.

Cost: no barge-in, and headphones for live demos.

Fully offline on one laptop

05

Because interview audio must not leave the device. It is decoded in a temp folder and discarded; recognition on Metal takes 0.73 s, not 2.3 s.

Cost: a 32 GB machine, and cold starts hidden by warm-up.

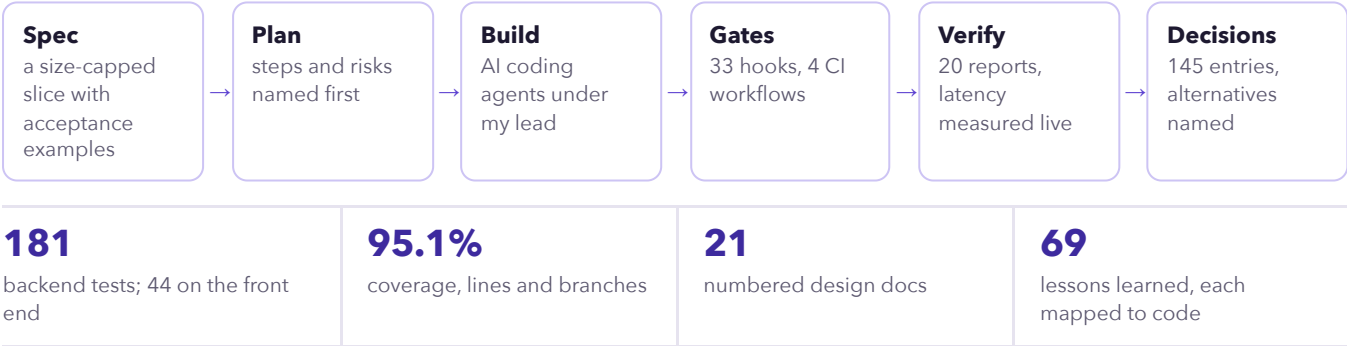
Latency is a release gate

06

Because unit tests cannot hear a pause. Every change to the voice path is re-measured on the real stack against a 3.0 s gate, stage by stage.

Cost: a live measurement run on every voice change.

HOW IT IS ENGINEERED



The contract is data, not prompt text: 21 rules across three files – English only, no scores, at most three sentences and 400 characters, and every turn ends with a question or a short acknowledgement. The loader rejects unknown checks; hard, soft and fallback outcomes are counted every turn.

WHERE IT TRANSFERS

Anywhere audio must not leave the device – clinical or legal intake, on-prem call-centre assistance, bank and insurance voice flows, kiosks, in-car systems, offline field tools. The shape carries over: a local speech loop that speaks sentence by sentence, a declarative contract checked before each sentence is voiced, and per-stage latency treated as a release gate. Each model sits behind a narrow interface, so moving to another model or a GPU server is re-wiring and re-measuring, not a redesign.

STACK

Backend Python, FastAPI with WebSocket, async SQLAlchemy, SQLite · **Speech** whisper.cpp on Metal, ffmpeg, Piper (ONNX) · **Models** Ollama: Qwen 2.5 instruct live, DeepSeek-R1 offline · **Frontend** React, TypeScript, Vite, Web Audio